NGUYEN QUANG HUY

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Summary

Senior Backend Engineer (Java) with 6+ years of experience building large-scale fintech and e-commerce platforms serving millions of users. Specialized in microservices and distributed systems: event-driven architecture on Kafka, cross-service data consistency (Transactional Outbox, idempotency), concurrency control, and Redis caching. Strong ownership of end-to-end architecture and system scalability, with a track record of leading cross-functional feature delivery, code reviews, and mentoring.

Work Experience

Arobid (B2B E-commerce) – Senior Engineer Jan 2026 – Present

- Building a new AI-powered B2B cross-border trade platform (NestJS, TypeScript) that combines B2B e-commerce with online trade expos, connecting suppliers with international buyers.
- **Identity & Onboarding:** Designed the authentication and user management system, including KYC (individual) and KYB (business) verification flows for onboarding buyers and suppliers.
- **Multi-tenancy:** Designed the platform's multi-tenant architecture on a shared schema with tenant-ID-based data isolation.
- **RFQ:** Designed the Request for Quotation (RFQ) flows between buyers and suppliers.
- **AI integration:** Integrated with the AI team's services to deliver AI-assisted onboarding and automated bulk onboarding of businesses from an uploaded list of company websites.
- Contributed backend APIs to the online trade expo module.

Inspire Lab Technology (Fintech) Sep 2021 – Oct 2025

- Designed and implemented core services of a microservices-based fintech platform:
  - **Identity & Security:** IAM Service migration (legacy Auth), Smart OTP, Risk Management.
  - **Payment & Transaction:** P2P Service, payment gateway integrations, transaction reconciliation.
  - **Reporting & Analytics:** Data pipelines and financial reporting dashboards.
- **Data consistency:** Implemented the Transactional Outbox pattern with idempotent consumers to keep data consistent across services, preventing duplicate processing under retries and at-least-once delivery.
- **Event-driven architecture:** Moved blocking synchronous service-to-service calls to asynchronous Kafka flows (partition keys for ordering, consumer groups for scaling), decoupling services and absorbing traffic spikes.
- **Concurrency control:** Prevented race conditions and lost updates on concurrent transactions using atomic conditional updates and optimistic/pessimistic locking in PostgreSQL, with explicit Spring transaction boundaries and propagation.
- **Caching & performance:** Introduced Redis caching for critical endpoints, using distributed locks to prevent cache breakdown on hot keys; tuned slow PostgreSQL queries.
- Built integration layers for third-party providers (OneSignal, SendGrid, Twilio, captcha services) behind a pluggable provider abstraction (Strategy pattern), so providers can be swapped without touching business logic.
- Applied Clean Architecture principles for improved modularity and maintainability.
- Collaborated with DevOps to improve GitLab CI/CD pipelines (automated tests and SonarQube quality gates).
- Mentored junior developers and led code review sessions, helping standardize coding practices.

Thi Truong Si – TTS Service JSC (E-commerce) Dec 2019 – Sep 2021

– Developed and maintained an e-commerce marketplace serving over 1,000,000 wholesale customers, including seller/buyer modules, RFQ, order processing, and payment features (MOMO, COD).

– Designed and implemented RESTful APIs with PHP (Laravel) and Golang, collaborating with frontend and mobile teams; optimized server-side code to improve response times and reduce latency.

– Designed and maintained database schemas with MySQL, MongoDB, and ArangoDB, implementing efficient data retrieval and manipulation operations.

– Built responsive web applications and admin dashboards with React.js (Redux, Axios), collaborating with UX/UI designers to deliver performant user interfaces.

Education

2016 – 2020 Bachelor of Software Engineering, Thu Dau Mot University

Technical Skills

Backend	Java, Spring Boot, NestJS (TypeScript), Golang, PHP (Laravel)
Architecture	Microservices, Event-driven Architecture, Multi-tenancy, Clean Architecture, DDD
Distributed Systems	Transactional Outbox, Idempotency, Distributed Locking, Eventual Consistency
Concurrency	Race Condition / Lost Update prevention, Optimistic & Pessimistic Locking, Transaction Management
Messaging	Kafka
Databases & Caching	PostgreSQL, MySQL, MongoDB, ArangoDB, Redis
Testing & Quality	JUnit, Mockito, SonarQube, GitLab CI/CD

Leadership & Soft Skills

- Technical ownership: system design, trade-off analysis, and cross-functional feature delivery.
- Mentoring junior developers and driving code review standards.
- Problem-solving, analytical thinking, and fast learning of new technologies.